The likelihood function is,

$$L(\theta_1, \theta_2) = \left(\frac{1}{\sqrt{2\pi\theta_2}}\right)^n \cdot \exp\left\{-\frac{\sum_{i=1}^{n}(x_i-\theta_1)^2}{2\theta_2}\right\}.$$

where $X_1, X_2, X_3, \cdots, X_n \overset{iid}{\sim} N(\theta_1, \theta_2)$,

and $x_1, x_2, x_3, \cdots, x_n$ are "realizations" of $X_1, \cdots, X_n$.
(or called "observations")

$$\frac{\partial \ln(L(\theta_1,\theta_2))}{\partial \theta_1} = ?$$

$$\frac{\partial \ln(L(\theta_1,\theta_2))}{\partial \theta_2} = ?$$

Log-likelihood function is,

$$\ell := \ln(L(\theta_1, \theta_2)) = -\frac{n}{2}\ln(2\pi\theta_2) - \frac{\sum_{i=1}^{n}(x_i-\theta_1)^2}{2\theta_2}$$

$$\frac{\partial \ell}{\partial \theta_1} = \underbrace{+\frac{1}{2\theta_2}}_{*} \cdot \sum_{i=1}^{n} \underbrace{2(x_i-\theta_1)}$$

Recall: "chain rule" of derivative calculation.

$$\frac{\partial \ell}{\partial \theta_2} = -\left(\frac{n}{2}\right)\cdot \underbrace{\frac{2\pi}{2\pi\theta_2}}_{*} + \frac{\sum_{i=1}^{n}(x_i-\theta_1)^2}{2}\cdot\frac{1}{\theta_2^2}$$

Estimation equation:

$$\begin{cases} \dfrac{\partial \ell(\theta_1, \theta_2)}{\partial \theta_1} \overset{\text{set}}{=} 0 \\ \dfrac{\partial \ell(\theta_1, \theta_2)}{\partial \theta_2} \overset{\text{set}}{=} 0 \end{cases}$$

$$\Rightarrow \begin{cases} \sum_{i=1}^{n} (x_i - \theta_1) = 0 \\ -\dfrac{n}{\theta_2} + \dfrac{\sum_{i=1}^{n} (x_i - \theta_1)^2}{\theta_2^2} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \hat{\theta}_1 = \sum_{i=1}^{n} x_i / n = \bar{X}. \\ \hat{\theta}_2 = \dfrac{1}{n} \cdot \sum_{i=1}^{n} (x_i - \hat{\theta}_1)^2 \\ \quad = \dfrac{1}{n} \sum_{i=1}^{n} (x_i - \bar{X})^2. \end{cases}$$

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